

- 2 (a) Fig. 2.1 shows the molecular structure of a triglyceride molecule.

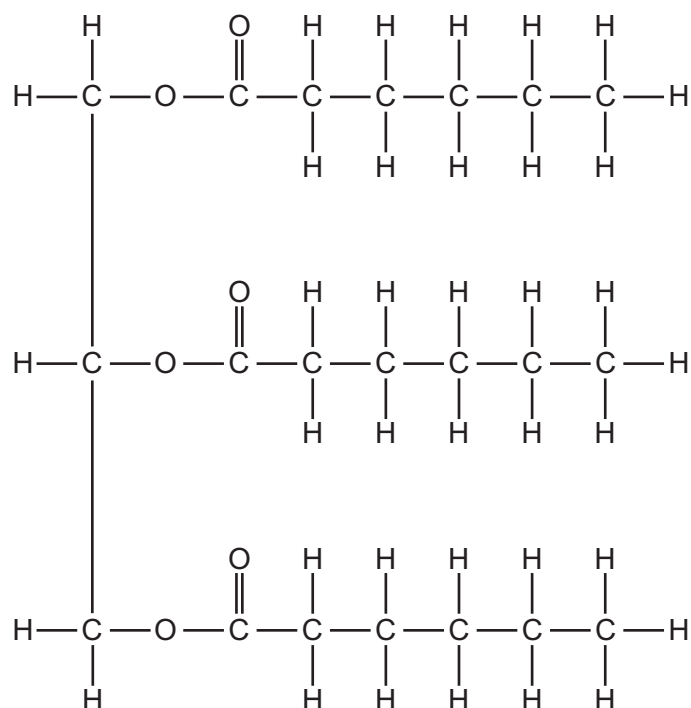


Fig. 2.1

- (i) Draw a circle around an ester bond shown in Fig. 2.1. [1]

- (ii) Name the type of reaction used to produce a triglyceride from its components.

State the number of water molecules produced during this reaction.

type of reaction

number of water molecules produced

[2]

(b) Lipases are enzymes that digest triglycerides in the lumen of the human intestine. These enzymes are released by exocytosis from intestinal epithelial cells.

(i) Underline **all** the terms from the list that are used to describe these lipases.

macromolecule

extracellular enzyme

fibrous protein

polysaccharide

[1]

Scientists have found that treating milk with lipase can improve its taste.

The scientists carried out an experiment to determine the effect of lipase activity on the triglycerides found in milk.

- Lipase was immobilised in alginate beads.
- The pH of a known volume of milk was adjusted to pH8 by adding an alkali.
- The beads were then mixed with this milk in a beaker.
- The pH of the reaction mixture was recorded over a period of 24 hours.

The results are shown in Fig. 2.2.

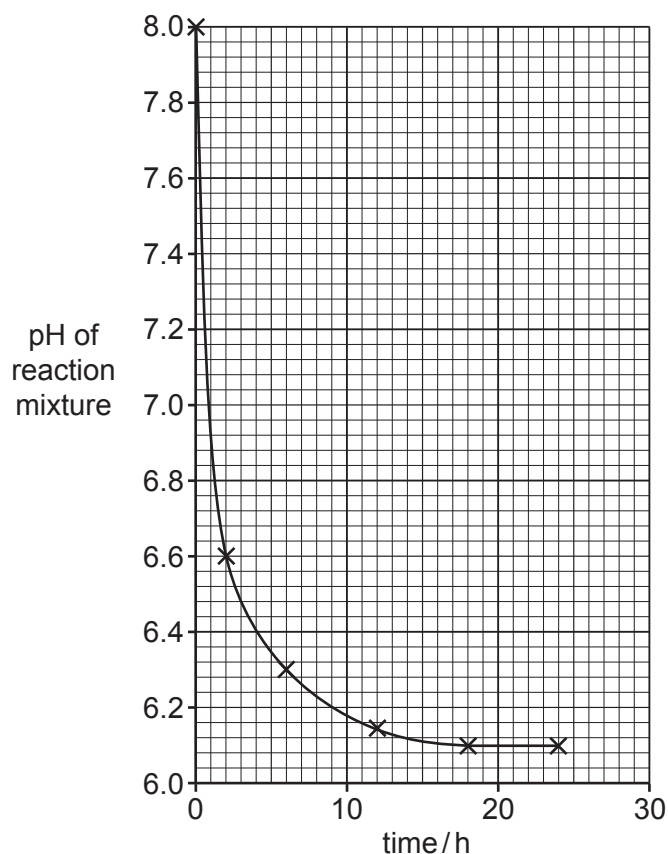


Fig. 2.2

(ii) Calculate the time taken for the pH to decrease from pH 6.6 to pH 6.3.

(iii) Explain the results shown in Fig. 2.2.

(iv) The scientists repeated the experiment using a higher concentration of lipase. All other variables remained constant.

[2]